

Stope and Pillar Room & Pillar

17/02/2020

Principle I

- *Stope and Pillar mining is oldest under ground mining method.
- *Sufficient large pillar is left in situ for support.
- *Roof support is also provided by leaving waste pillar.
- *Width of the stope (opening) is minimized the possibility of Roof Fall.
- *Basic principal of Stope & Pillar mining remains same. Changes are (1) introduction of machines (2) extending method to accommodate vertical expansion in massive deposit.

Principle II

*Stope and Pillar mining is also known as breast stoping (horizontal drilling at chest level). Stall & Pillar method (stall as additional support in stope).

Oldest Method of Working 1

- *Stope & pillar method is generally applicable in horizontal or near horizontal ore body. * In case the deposit has out crop on surface, the easy access has the advantages of early operation of ore body by stope & pillar method of min
- *Stope & pillar method of work can be executed without any additional support material.
- *Some amount of extra care for on pillar reduction can keep the stope back stable.

Oldest Method of Working 2

- *Ore can be taken out by wheel cart or head load, requiring no sophisticated additional machinery.
- *Soft ore can be cut by axe or pick axe, which are readily available in house hold.
- *Stope & pillar method is simplest mining method.
- *Stope & pillar method can be operated with low capital.
- *Selective mining is possible, low grade ore is left in pillars.
- *Ore Production is by development, stope starts producing ore as soon as ore body is reached.

Oldest Method of Working 3

- *Ore production can be varied by changing man power deployed in stope.
- *Very low capital investment.

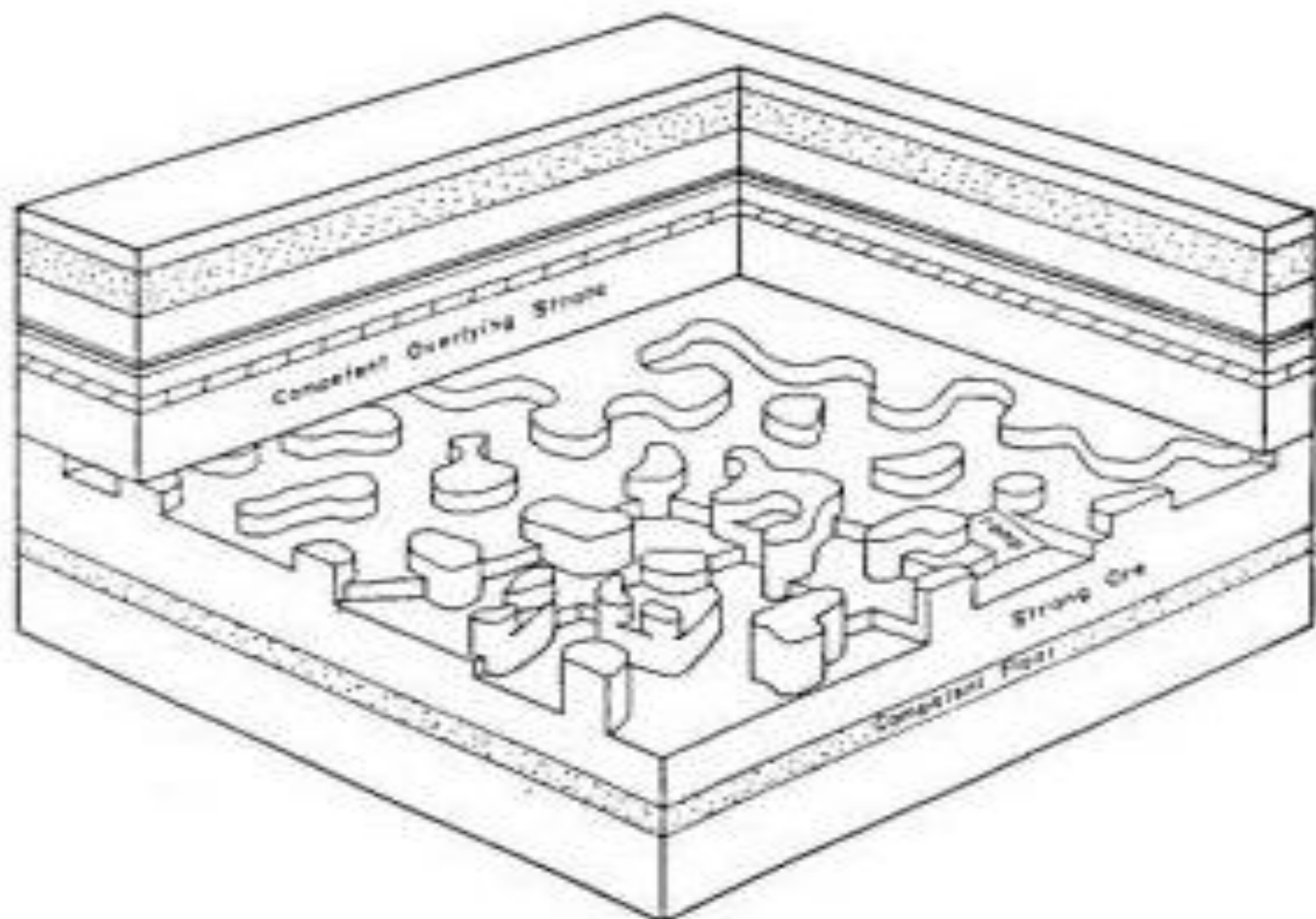


Fig. 18.2.1. Stope and pillar mining with random pillars (Clark and Gaudin, 1961)

General Feature I

- *Ore pillars are left in the stope.
- *Size of pillars are decided based on geo-mechanical properties of ore / formation.
- *Pillars are basically designed to support roof for safe working till end.
- *Pillar height is a major factor for deciding size of pillar.
- *Stope & Pillar is applied to flat ore body or dipping less than the angle of repose.

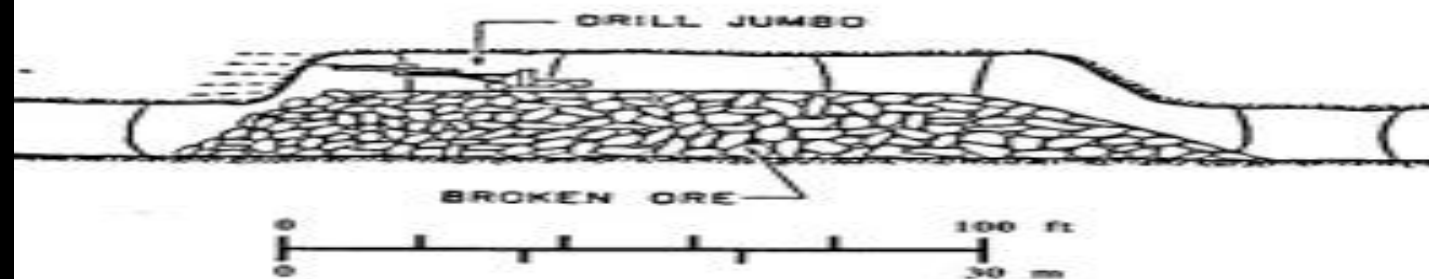
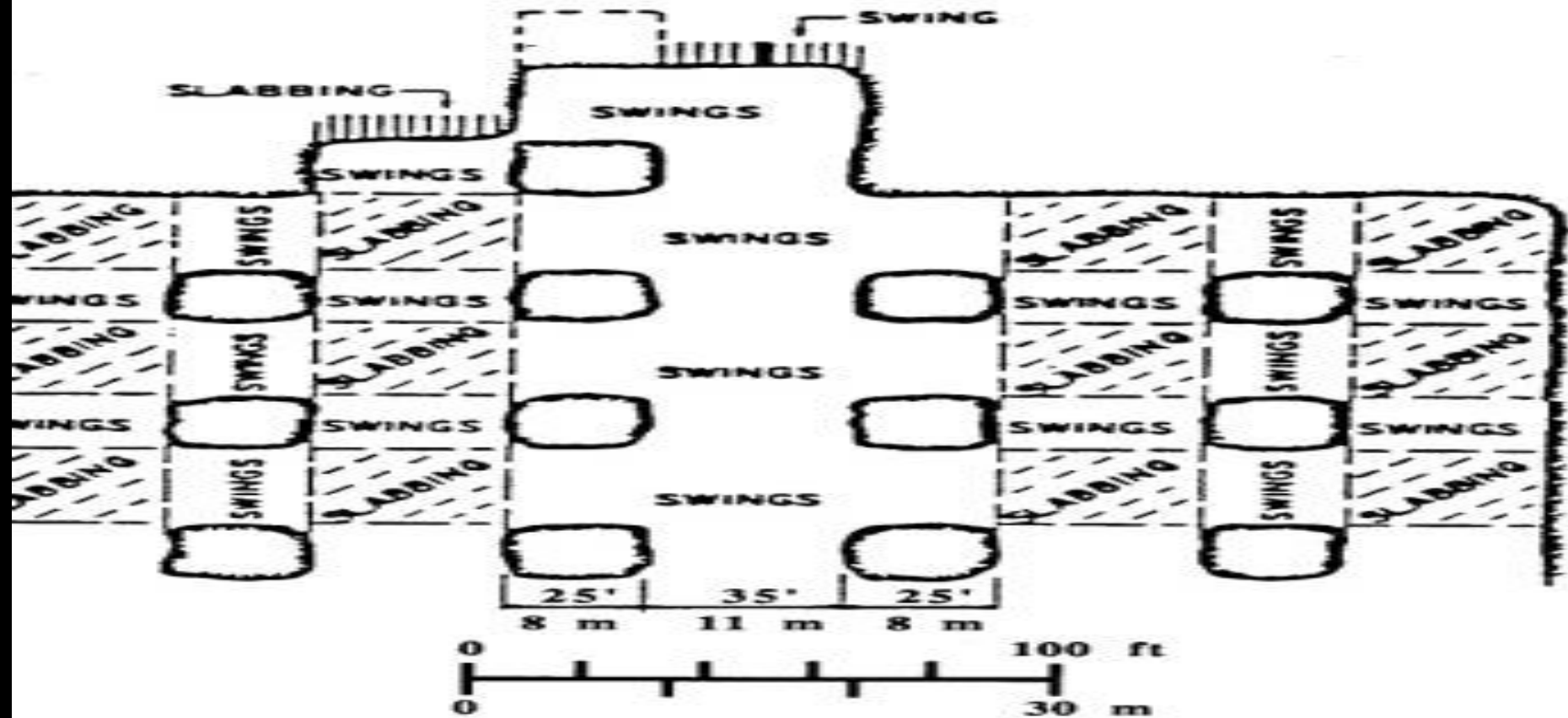
General Feature II

- *Maximum permissible dip is 40 degree for stope & pillar method of mining.
 - *Very thick ore body can be mined by Stope & Pillar method. By cutting horizontal slice either from top to bottom or bottom to top (upper hand or under hand).
 - *Competent rock mass is very desirable as this is an open method.
 - *Stope normally stay open during the life of panel.
- Desirable compressive strength of rock is 25MPa or above.

General Feature III

- *Structural competency of roof, pillars and floor components is important.
- *In case the structural competency is not very good, secondary support and frequent scaling becomes essential.
- *Uncontrolled pillar failure is to be avoided at any cost.
- *Pillar failure can be progressive to surrounding pillars.
- *To prevent the collapse of entire mine should be divided in different panels with barrier pillars.

STOPE AND PILLAR MINING

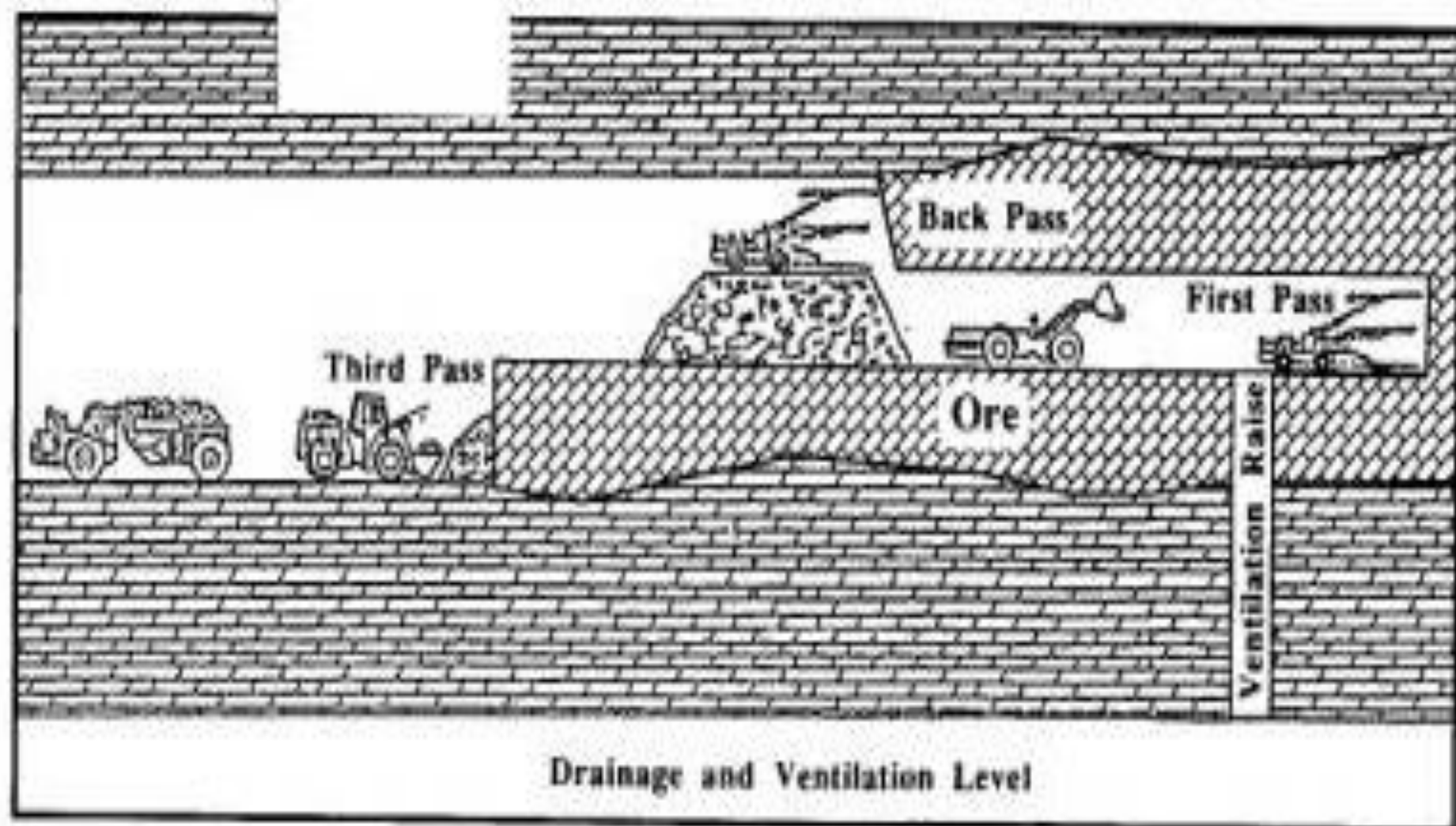


18.2.4. Overhand stope and pillar mining (Bullock, 1982).

18.2.2.6 Water

Water problems as mining operations are driven underground, the ore can be pumped away from the mine. Locally powered pumps are selected to fit the arrangements. Significant variations in water levels must be controlled.

STOPE AND PILLAR MINING



Disadvantages I

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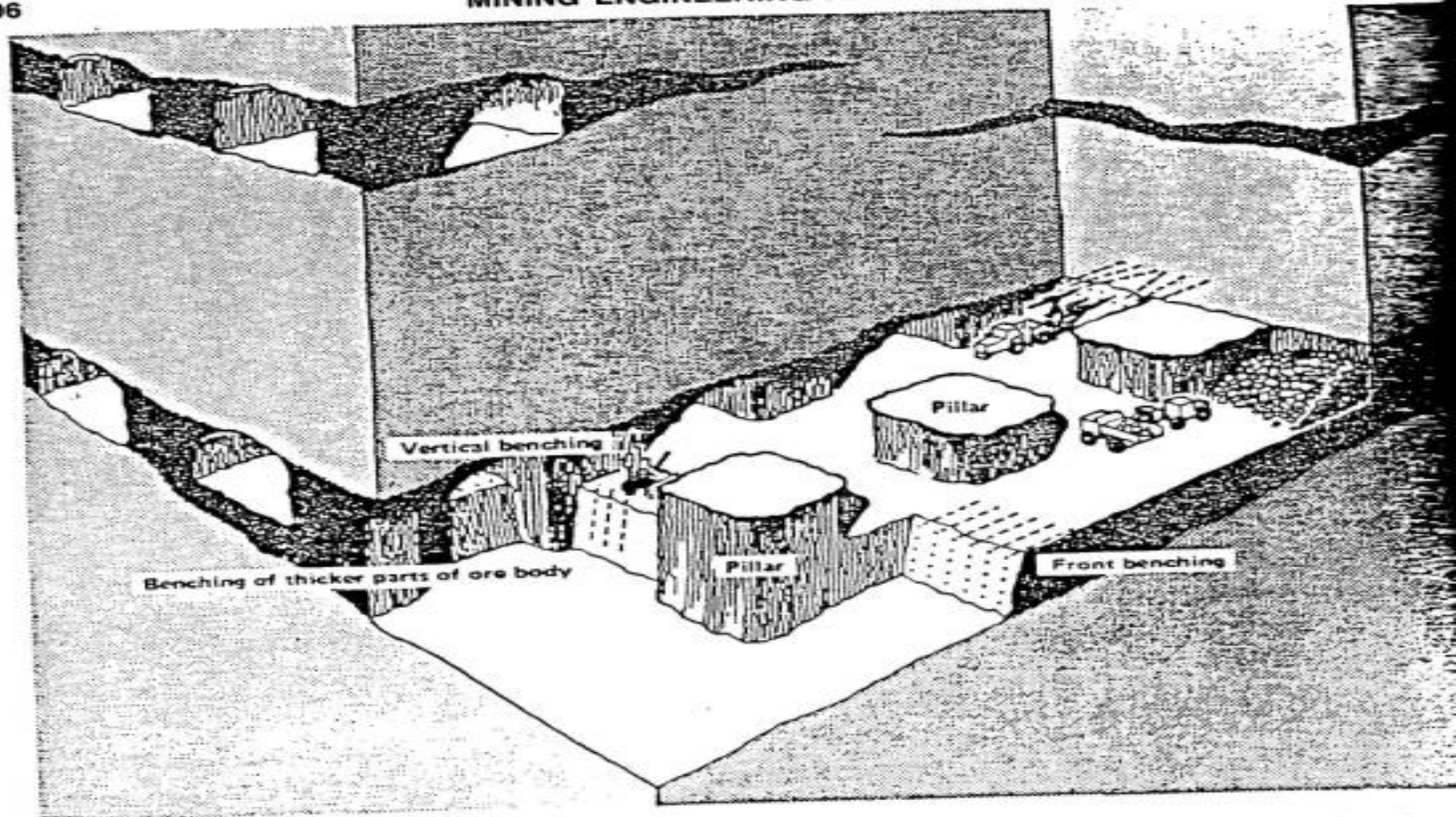


Fig. 18.2.5. Stope and pillar mining using vertical and front benching. (Hamrin, 1980. By permission: Atlas Copco.)

typically up to 500 ft (150 m), although production is

General Feature IV

- *Depth limitation is defined by load bearing capacity of pillars.
- *Increasing pillar size is counter productive because of decrease in the extraction.
- *Typical depth limitation of Stope & Pillar method of mining is 600 meter.
- *Surface subsidence is absent expect when massive roof or pillar failure takes place.
- *Stope & Pillar is large scale method of mining and flexible enough for wide range of production requirement.

Development I

- *Stopes are developed by massive development program.
- *Drive (along the strike) and raise (along the dip) is excavated at suitable interval.
- *Incase of irregular ore body or a very massive extensive ore body, connecting ore pass, ventilation ways, access raise are necessary.
- *Ore hoisting mostly take place from one point by vertical shaft.
- *decline for movement of rubber mounted underground equipment.

Stoping I

- *Stoping operation is cyclic in nature.
- *Stoping includes stripping of pillars to final size.
- *At the ultimate size of pillars, the stope stands, till end mining of the panel is completed.
- *Roof is allowed to collapse allowing subsidence at surface.
- *Stopes can be filled up by filling to prevent subsidence at surface.
- *Cycle of operation for hard ore is Drilling >Blasting >Dressing > Support >Loading >Transport >Dump.

Stoping II

- *Cycle of operation for soft ore is Cutting (by machine) >Dressing (Minor) >Support >Loading >Transport >Dump.
- *Drilling is carried out by Electro Hydraulic Drill Jumbo.
- *In of small deposit, hand held jack hammers used.
- *Very soft rock is cut by contiguous cutter or augur drill.
- *Wedge cut, V Cut is preferred for controlling through after blasting.
- *Burn cut is also practiced in hard rock.

Stoping III

- *Face up to 9m in height is taken with hole depth up to 4.8 m.
- *Predominating explosive is Emulsion, Cartridge or site mixing explosive based on the scale of operation.
- *Blasting is generally carried out in between shift.
- *Good fragmentation plus limiting throw of round is important.
- *Clearing large area and secondary blasting disrupts mining process.
- *For dressing back and pillar scalar is used.

Stoping IV

- *Stalls (wooden props) erected (whenever required) long with advancement of face.
- *For ore body more than 9 m thick, is mined by multiple pass system.
- *First pass is taken at lower portion of the ore body.
- *First pass defines ore body and pillars.
- *First pass is taken from lower side to upwards (up direction).
- *Topmost pass is than mined out by using the broken ore as platform.

Stoping V

- *Once mining is complete , broken ore is removed at one time.
- *Stock pile of broken rock blocks considerable capital of the mine.
- *Under lying pass can be mined by vertical (by wagon drill) or front benching (by drill Jumbo)

Pillar Design 1

- *Ore pillars are left in the stope, strength of ore is very important in stope & pillar stope.
- *In case the structural competency of pillar is poor, secondary support and frequent scaling of pillar is required.
- *Size of pillars are decided based on geo-mechanical properties of ore / formation.
- *Pillars are basically designed to support roof for safe working till end.
- *Pillar height is a major factor for deciding size of pillar.

Pillar Design 2

- *Pillar design has to take care of dip of the ore body.
- *Pillars are supported on foot wall of stope, strength of foot wall rock is also important to design of pillar.
- *Pillars are normally designed to last life time of the stope.
- *Desirable strength of rock mass for pillar is 25MPa.
- *Uncontrolled pillar failure is to be avoided at any cost at design stage itself.

Pillar Design 3

- *Pillar failure can be progressive to surrounding pillars.
- *To prevent the collapse of entire mine should be divided in different panels with barrier pillars.
- *Size of blasting, heavy blast creates more vibration & can affect ore pillars, which is not structurally sound.
- *Depth of pillar (depth of ore body)
- *Water bearing rock (HW, FW) or water bearing ore may damage strength
- *Pillars with treatment.
- *

Pillar With Treatment 1

- *Under multiple horizon mining, the pillars are wrapped mainly after stripping.
- *Wrapping is done by using wire ropes, encasing pillar in shot crete.
- *This system enhances strength of the pillar and increase the resistance to shear failure.
- *Wrapping is done, along with Horizontal Rock Bolt.
- *For monitoring pillars, instrumentation can be done to know the health of pillar.

Pillar Recovery I

- *Pillar recovery improves extraction.
- *Pillars are recovered at the end of the life of the panel.
- *Pillars are recovered in retreating mode (back to front).
- *After recovery of pillar, the back of the stope cave and swell.
- *Swelled rock support the Rock over head.
- *Wherever , caving is not permissible, the stope is filled up by suitable filling material.

Loading & Haulage I

- *Track less Load Haul Dump (LHD) machines are used for loading a Low Profile Dump Truck (LPDT).
- *Gathering arm loader (Slusher) are some time used for loading trucks (LPDT)
- *Long hauling for large quantity of ore is cheaper by Rail Mounted car or Belt Conveyor.
- *Rail car or Belt conveyers are normally placed at foot wall drift.
- *Foot wall drift allows a stable passage and with fair gradient.

Ground Control I

- *Ground controls is taken care by Geotechnical studies of ore, hang wall and foot wall formation.
- *Pillar size, room size, panel size, barrier pillar size are designed before the mining operation start by numerical method.
- *Scaling of loose rock is done by scalar where ever required.
- *Under multiple horizon mining, the pillars are wrapped after stripping.
- *Wrapping is done by using wire ropes, encasing pillar in shot crete.

Ground Control II

- *This system enhance strength of the pillar and increase the resistance to shear failure.
- *Wrapping is done, along with Horizontal Rock Bolt.
- *Pillar failure develops the condition of a pressure arching.
- *Surrounding pillars are subjected to high stress, causing more failure.
- *Panel barrier pillar should be sufficiently strong to stand the pressure arching.
- *Pressure arches, if not controlled, can cause entire mine to collapse.

Ventilation I

- *Large volume of stope makes the ventilation air flow slow.
- *Large volume of air normally dilutes exhaust gas.
- *In case of presence of any other gas, air flow has to be increased by booster fan.
- *Air stopping's are some time used to increase air flow near active faces.

Water Pumping I

- *For incline ore body, water sump is created at the deepest point for pumping out water.
- *For handling large quantity of water, pumping from intermittent sump proves cheaper.
- *In case of flat (tabular) ore body, smaller intermediate sump with floating pump shall be more use full for dewatering.

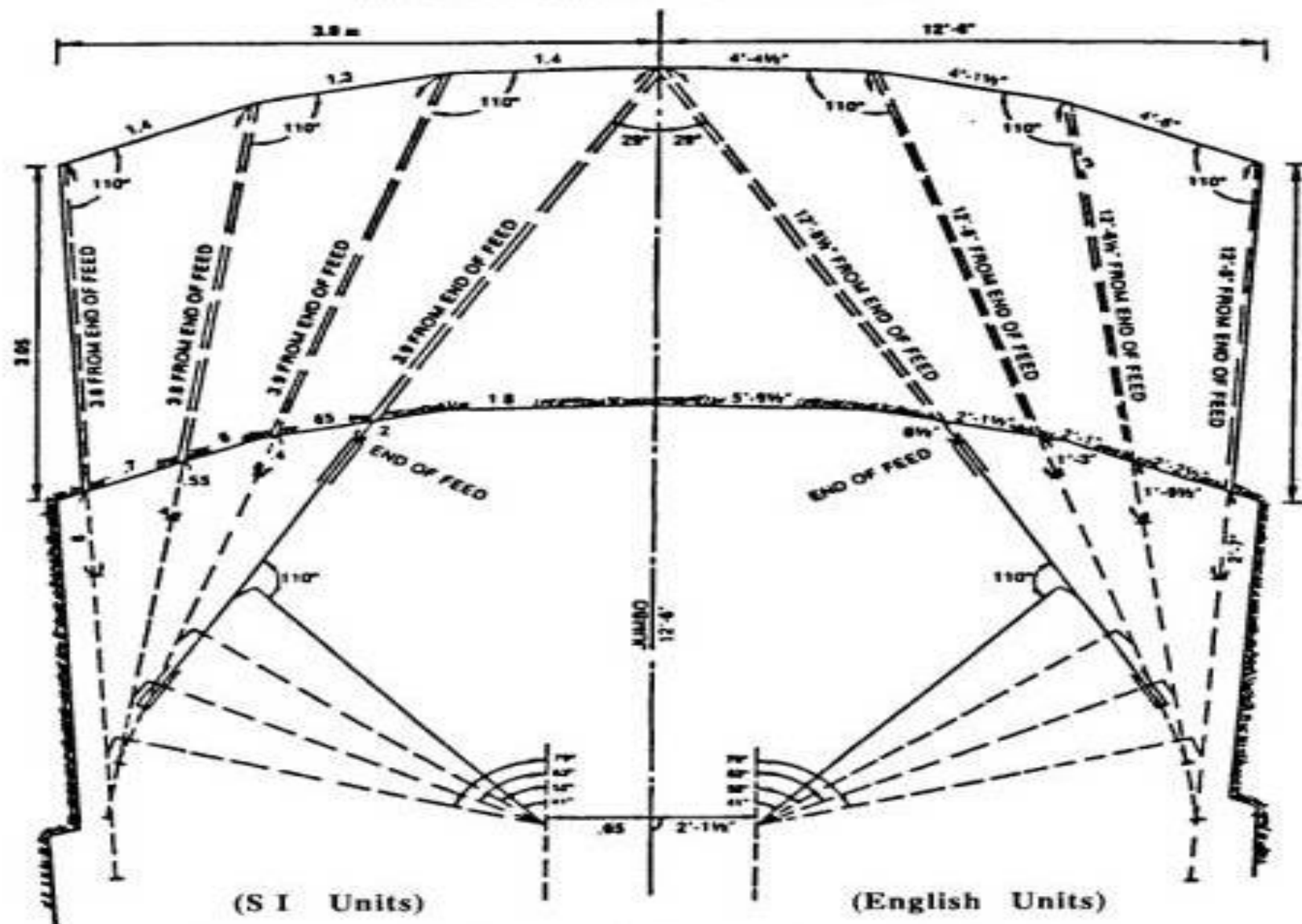
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Fig. 18.2.2. Typical V cut as used in dolomites or limestones (Bullock, 1982).

Incline Stope & Pillar I

- *For the deposit in between 10 to 30 degree dip, incline Stope & Pillar is adopted.
- *Lower level also used as haulage level is equipped with rail track and loading arrangement.
- *Upper level, normally 30 meter above lower level is excavated and used as access level.
- *Raises are taken manually from lower level to upper level.
- *Scraper or slushier are installed in the raise to scarp down blasted ore to lower level draw point.

Incline Stope & Pillar II

- *Raises are stripped from bottom to top by drilling hole with jack hammer.
- *Rib pillars are left in between two adjacent rooms as per stope design.
- *Rock bolts are extensively used for roof and pillar support (where required).
- *Broken ore is loaded by loader in to mine car for transportation.

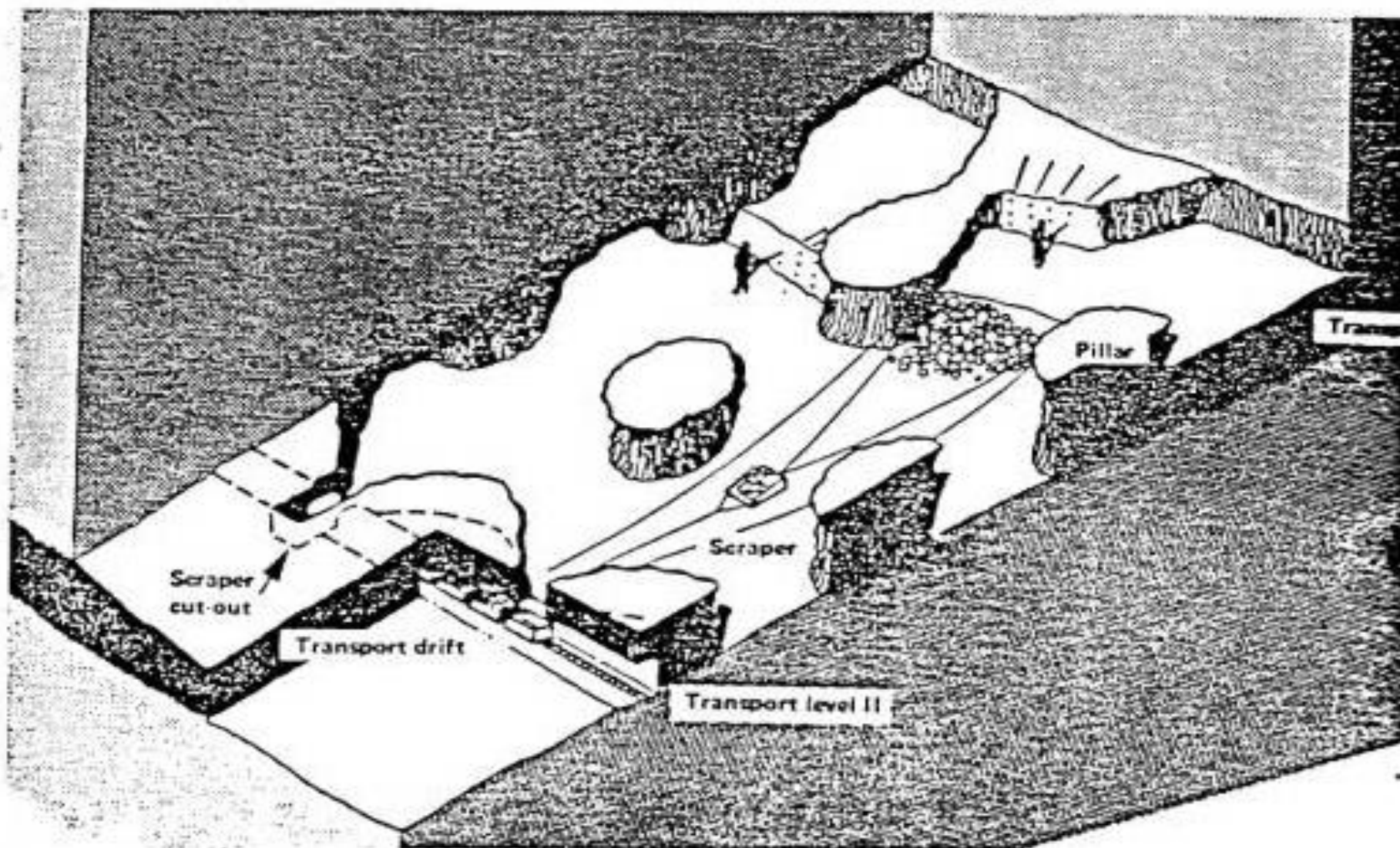


Fig. 18.2.7. Inclined stope and pillar operation in inclined ore using slushers and hand-held drills. (Hamrin, 1980. By permission of Atlas Copco.)

Step Stope & Pillar I

- *This method is followed for the deposit below 10 degree dip.
- *Step stope pillar mining utilizes trackless equipment for mining.
- *Set of drive is taken parallel to strike direction of the ore body.
- *Drive / Raise along the dip direction is taken at apparent dip angle, so that the trackless machine can operate at this angle.

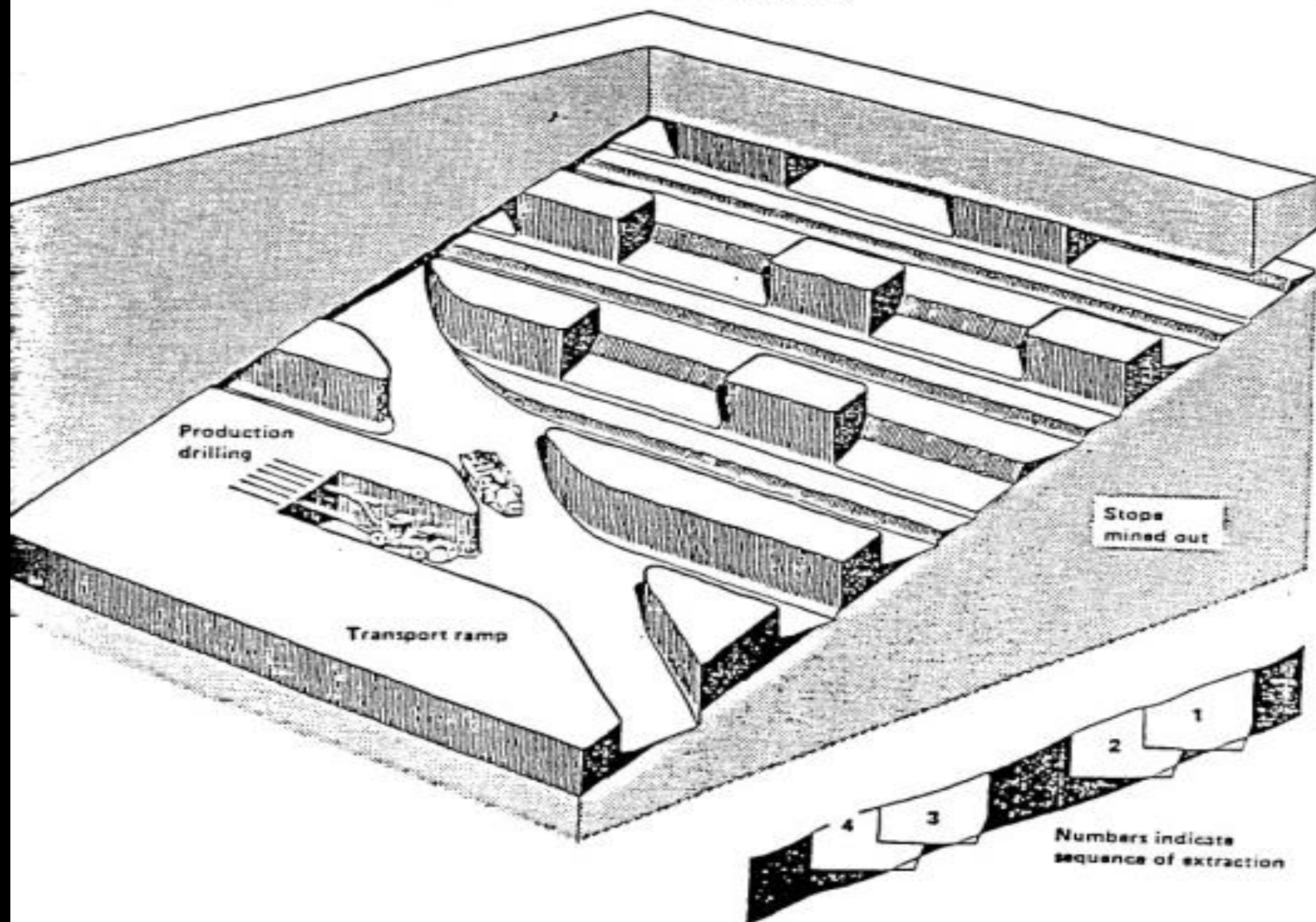


Fig. 18.2.6. Step stope and pillar in an inclined ore body. (Hamrin, 1980. By permission: Atlas Copco.)

Advantages I

- *Method can be mechanized
- *Utilization of equipment can be computerized for optimizations.
- *Method can be selective by leaving low grade ore in pillars
- *Requirement of development before production is low.
- *Higher production can be achieved by opening multiple faces.
- *High capacity trackless production machines can be deployed.

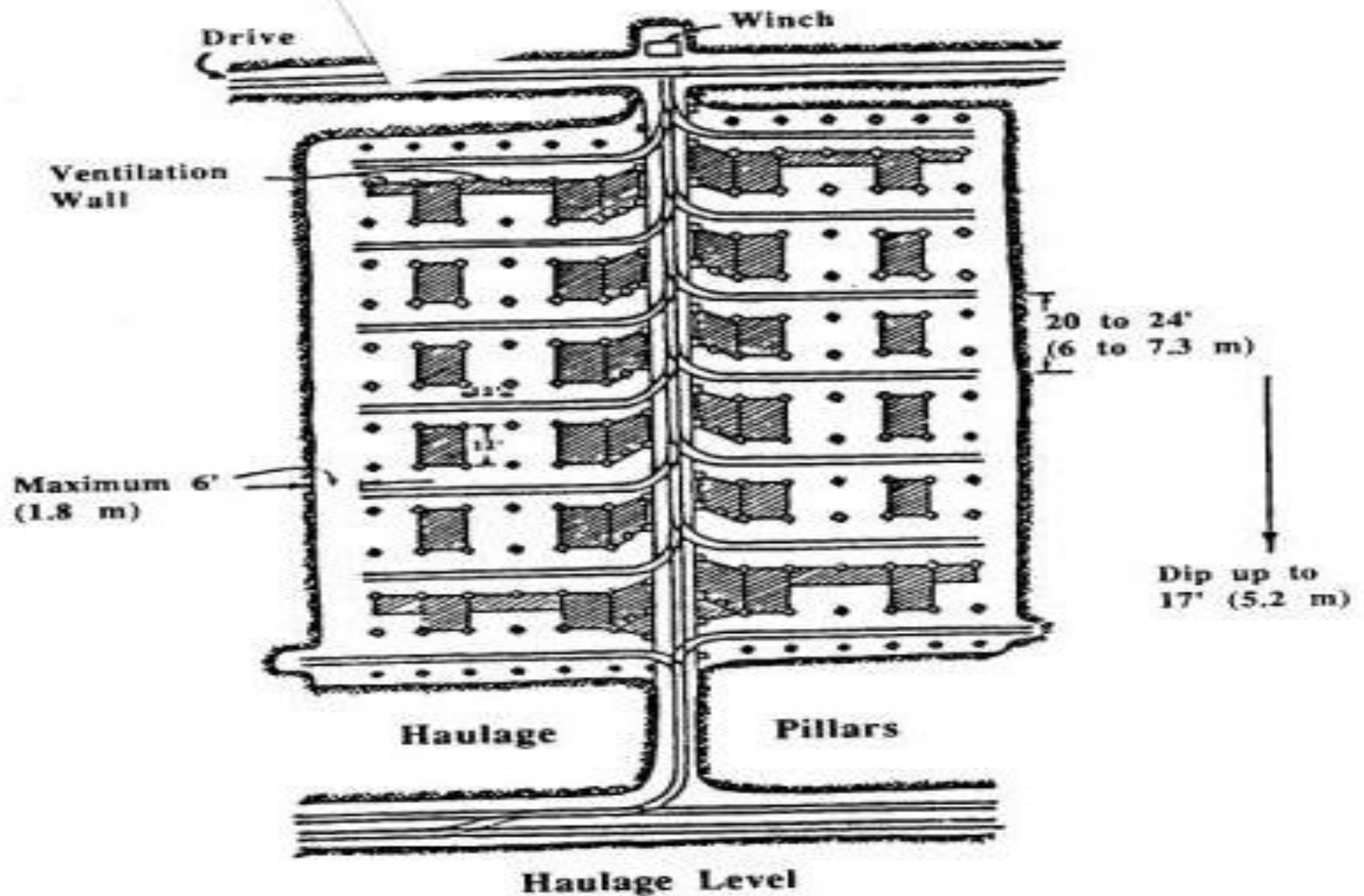
Advantages II

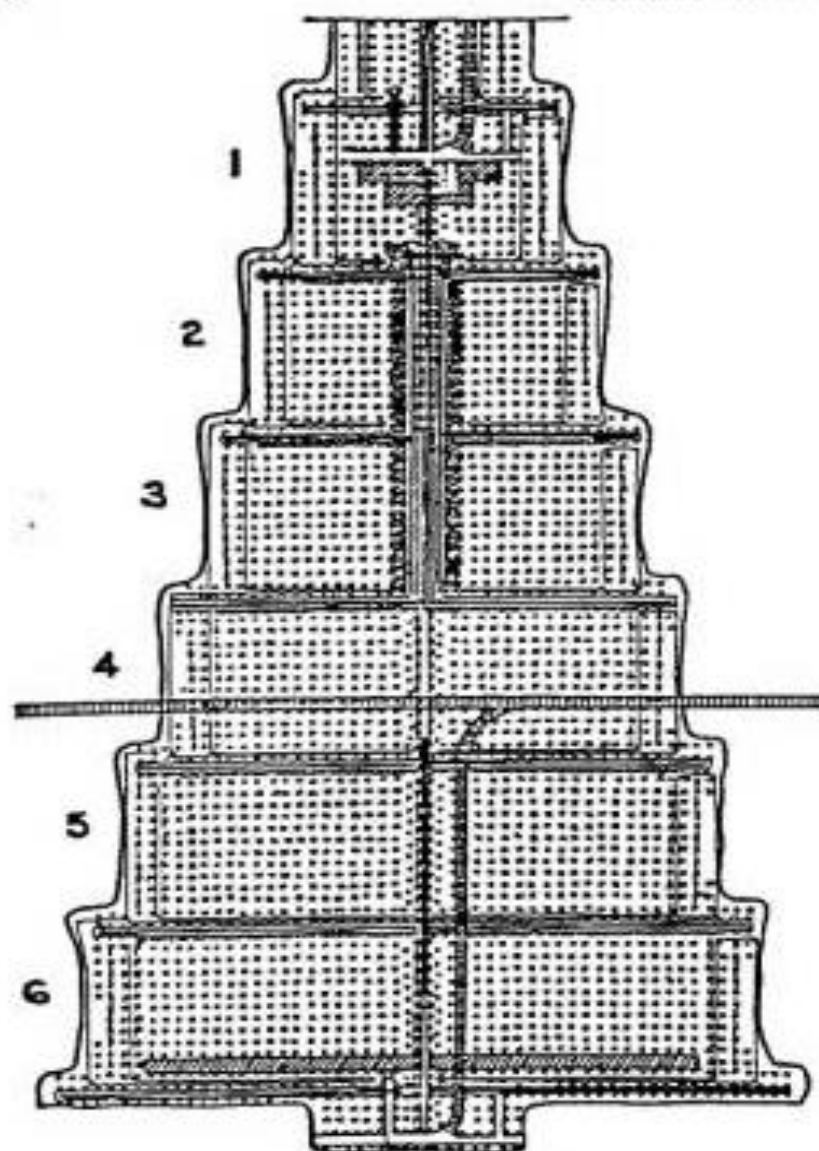
- *Method is not labour intensive.
- *Production rate is 50-70 t per man shift (OMS).
- *Dilution is normally low.
- *Surface subsidence can be avoided, if suitably planned.

Disadvantages II

- *Low recovery, up to 30% ore or more likely to be locked up in pillar.
- *Stope ventilation is difficult.
- *Maintaining open space in underground mine is costly for bad ground condition.
- *Water may create problem while mining flat ore body, particularly undulating deposit.

STOPE AND PILLAR MINING





Dip 25°; Length of longwall 900 ft (275 m); Stopping width 39 in. (1 m)

Fig. 18.2.9. Longwall breast stoping operation. (Beck et al., 1961.
By permission: South African Institute of Mining and Metallurgy.)

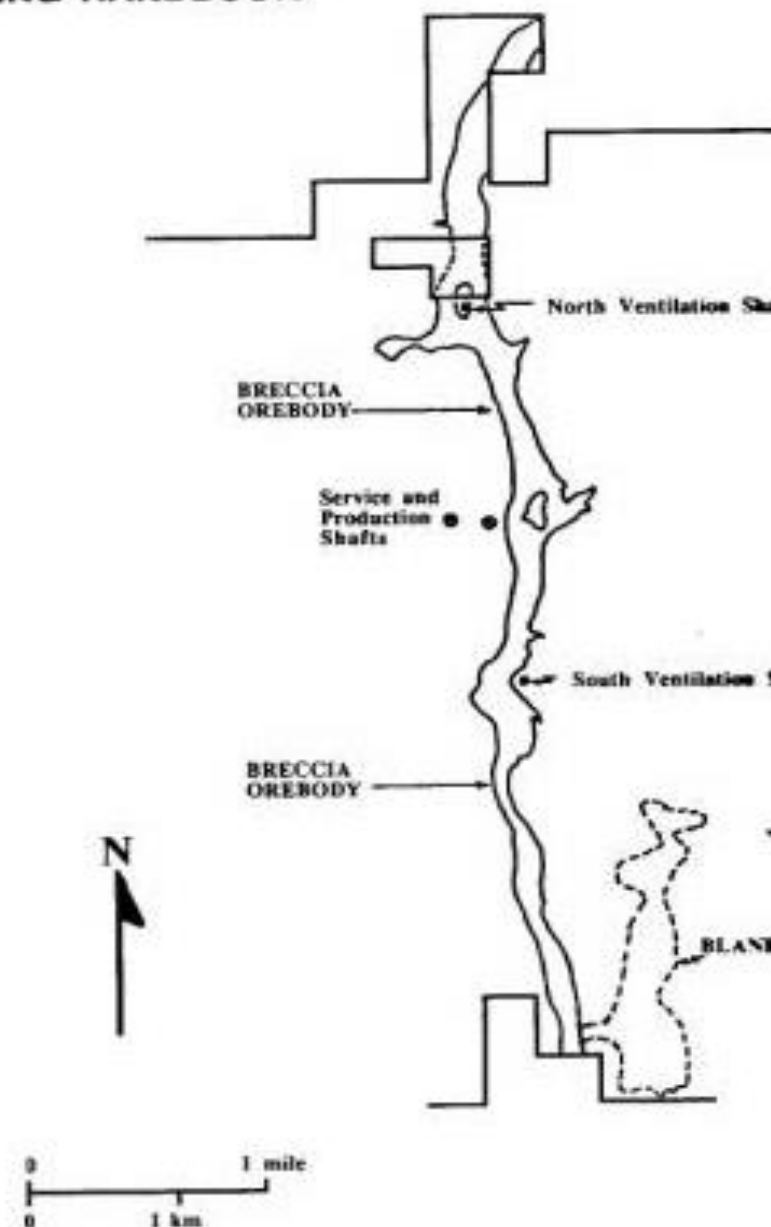


Fig. 18.2.10. Buick mine ore body (Anon., 1961.)

Reference

*SME Mining Engineering Handbook, Hartmen, 2nd
Edition, Volume 2, page 1702 to 1711